QI LIU

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EDUCATION

Renmin University of China, Gaoling School of Artificial Intelligence — Sep 2024–present
PhD student in Artificial Intelligence; advisor: Prof. Jiaxin Mao — Beijing, China

Renmin University of China, Gaoling School of Artificial Intelligence — Sep 2020–Jun 2024
BEng in Artificial Intelligence — Beijing, China

RESEARCH DIRECTIONS

I. LLM for Search

Multi-vector retrieval. My early *model analysis* and *matching and pruning study* explored how multi-vector retrieval balances effectiveness and efficiency. I also led development of the open-source long-context late-interaction retriever jina-colbert.

Dense retrieval models. From the bilingual jina-embeddings-v2 model, I moved toward broader uses of embeddings: *E2Rank* extends them to listwise reranking, and *Learning to Retrieve* explores reinforcement learning in embedding space.

LLM for reranking. This line spans efficient listwise ranking *with passage embeddings*, reusable methods in *LLM4Ranking*, *relevance interpretation*, and the diffusion-based approach *DiffuRank*.

II. Search Agents

Behavior and information acquisition. I analyze how long-horizon agents search and where they fail in *Diagnosing Search Behavior*; *Fetch-then-Explore* then separates source selection from extraction over a persistent workspace.

Search as Code (ongoing). I am developing *OpenSaC*, a Python interface that lets deep-research agents organize and execute search through code.

TECHNICAL SKILLS

Coding: Python, Linux, and Git; working knowledge of C/C++, SQL, and Bash. Proficient with Codex and Claude Code for implementation, debugging, and research prototyping.

AI frameworks: PyTorch, Hugging Face Transformers, DeepSpeed, vLLM, slime, and Pyserini for model training, inference, and retrieval; Harbor for agent evaluation.

RESEARCH AND INDUSTRY EXPERIENCE

Xiaohongshu — May 2026–present
Research intern — Beijing, China

NExT++ Research Center, National University of Singapore — Jan–Jul 2026
Visiting PhD student; worked with Dr. Fengbin Zhu and Prof. Tat-Seng Chua on search agents. — Singapore

Tongyi Lab, Alibaba Group — May 2025–Jan 2026
Research intern; worked on text embedding and reranking models. — Beijing, China

Jina AI — Nov 2023–Jun 2024
Research intern; developed embedding and retrieval models — Beijing, China

Xiaohongshu — Jul–Sep 2023
NLP algorithm intern — Beijing, China

SELECTED PUBLICATIONS

1. **OpenSaC: A Python Library for Search as Code in Deep Research Agents**
   Qi Liu, Jiaxin Mao
   Preprint, 2026.

2. **Learning to Retrieve via Reinforcement Learning in Embedding Space**
   Qi Liu, Fengming Liang, Yiqun Chen, Erhan Zhang, Jiaxin Mao
   Preprint, 2026.

3. **Fetch-then-Explore: Decoupling Selection from Extraction over a Persistent Workspace for Search Agents**
   Qi Liu, Yiqun Chen, Zidan Chen, Yan Gao, Yi Wu, Yao Hu, Jiaxin Mao, Fengbin Zhu, Tat-Seng Chua
   Preprint, 2026.

4. **Diagnosing Search Behavior and Failure Modes in Long-Horizon Search Agents**
   Qi Liu, Jiaxin Mao, Fengbin Zhu, Tat-Seng Chua
   SIGIR-AP 2026.

5. **DiffuRank: Effective Document Reranking with Diffusion Language Models**
   Qi Liu, Kun Ai, Jiaxin Mao, Yanzhao Zhang, Mingxin Li, Dingkun Long, Pengjun Xie, Fengbin Zhu, Ji-Rong Wen
   Preprint, 2026.

6. **E2Rank: Your Text Embedding can Also be an Effective and Efficient Listwise Reranker**
   Qi Liu, Yanzhao Zhang, Mingxin Li, Dingkun Long, Pengjun Xie, Jiaxin Mao
   EMNLP 2026.

7. **How do Large Language Models Understand Relevance? A Mechanistic Interpretability Perspective**
   Qi Liu, Haozhe Duan, Jiaxin Mao, Ji-Rong Wen
   ACM Transactions on Information Systems, 2025.

8. **LLM4Ranking: An Easy-to-use Framework of Utilizing Large Language Models for Document Reranking**
   Qi Liu, Haozhe Duan, Yiqun Chen, Quanfeng Lu, Weiwei Sun, Jiaxin Mao
   Preprint, 2025.

9. **Leveraging Passage Embeddings for Efficient Listwise Reranking with Large Language Models**
   Qi Liu, Bo Wang, Nan Wang, Jiaxin Mao
   WWW 2025.

10. **An Analysis on Matching Mechanisms and Token Pruning for Late-interaction Models**
    Qi Liu*, Gang Guo*, Jiaxin Mao, Zhicheng Dou, Ji-Rong Wen, Hao Jiang, Xinyu Zhang, Zhao Cao
    ACM Transactions on Information Systems, 2023.

11. **Understanding the Multi-vector Dense Retrieval Models**
    Qi Liu, Jiaxin Mao
    CIKM 2023.